

Common Error Summary and Solutions Documentation

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I. multi_brains_pre Package

1.1 Configuration File Related Errors

Error 1.1.1: Configuration file not found

Error Log:

```
multi_brains_pre_setting.yaml file not found, Please check: whether the file exists: {config_file}
```

Cause:

- The path for the `multi_brains_pre_setting.yaml` configuration file does not exist.
- The file has been deleted or moved.

Solution:

1. Copy `multi_brains_pre_setting.yaml` from the attached source code files to the specified directory.
-

Error 1.1.2: YAML Parsing Failed

Error Log:

```
multi_brains_pre_setting.yaml file YAML parsing failed: {exc}, Please check:  
whether the file format has been modified
```

Cause:

- Incorrect YAML file format (e.g., incorrect indentation, unescaped special characters).

Solution:

1. Modify `multi_brains_pre_setting.yaml`, referencing the default file format found in the attached source code files.

1.2 ASR Speech Recognition Errors

Error 1.2.1: ASR Engine Initialization Failed

Error Log:

```
ASR Engine: {model_name} initialized failed
```

Cause:

- Incorrect or invalid API Key configuration.
- Unsupported model name.
- Network issues (for online ASR services).
- Key file decryption failed (for iFlytek ASR).

Solution:

1. Check if `ALIYUN_API_KEY` is configured correctly.
2. Verify that the model name being used is included in the supported list:
 - paraformer-realtime-v2/v1
 - paraformer-realtime-8k-v2/v1
 - gummy-realtime-v1/gummy-chat-v1
3. For iFlytek ASR, check if the key file path and decryption settings are correct.
4. Test your network connection.

Error 1.2.2: Speech Recognition Failed

Error Log:

```
asr recognize failure  
online ASR recognition failed: {error_message}
```

Cause:

- Invalid or expired API Key.
- The online speech recognition service is currently unavailable.

Solution:

1. Check the stability of your network connection.
 2. Check if your API usage quota has been exhausted.
-

Error 1.2.3: VAD Engine Initialization Failed**Error Log:**

```
VadEngine initialization failed  
VadEngine initialization exception: {exc}
```

Cause:

- The VAD model file does not exist.
- The model path is configured incorrectly.
- The `voice_toolbox` library was not installed correctly.

Solution:

1. Check if the `silero_vad.int8.onnx` model file has been deleted.
 2. Download the VAD model file from the attached source code.
-

Error 1.2.4: Audio Stream Open Failed**Error Log:**

```
Audio stream open failed: {exc}
```

Cause:

- The microphone device is not connected.
- The audio device is currently in use by another program.

Solution:

1. Check if the microphone device is connected correctly.
 2. Use `arecord -l` to list available audio devices.
 3. Close any other programs that are currently using the audio device.
-

Error 1.2.5: No Speech Detected**Warning Log:**

```
No speech detected, timeout exit.
```

Cause:

- Microphone volume is too low.
- Ambient noise is excessive.

Solution:

1. Increase the microphone input volume.
2. Reduce ambient noise.

3. Speak closer to the microphone.

Error 1.2.6: Unsupported Language

Warning Log:

```
Unsupported language: {language}, only 'zh' and 'en' are supported
```

Cause:

- An unsupported language code was configured.
- The `LANGUAGE` parameter is set incorrectly.

Solution:

1. Change the `LANGUAGE` parameter to 'zh' (Chinese) or 'en' (English).
2. Check the language settings within the configuration file:
`$HOME/M3Pro_ws/src/m3pro_bringup/config/multi_brains_pre_setting.yaml`

1.3 TTS (Text-to-Speech) Related Errors

Error 1.3.1: TTS Engine Initialization Failed

Error Log:

```
Failed to initialize Piper TTS engine: {e}  
Failed to initialize Baidu TTS engine: {e}  
Tongyi TTS_Engine initialized failed  
Baidu TTS_Engine initialized failed  
Xunfei TTS_Engine initialized failed  
Piper TTS_Engine initialized failed
```

Cause:

- TTS model files are missing.
- API Key is configured incorrectly.
- Configuration file failed to load.
- Key file decryption failed (Xunfei TTS).

Solution:

1. Piper TTS:

- Check if the model files exist in the `~/MODELS/tts/` directory.
- Confirm that both the `.onnx` and `.onnx.json` files are present.

2. Online TTS (Alibaba Cloud/Baidu/Xunfei):

- Check if the corresponding API Key is configured correctly.
- Verify the network connection.

3. Check if the configuration file format is correct.
-

Error 1.3.2: Speech Synthesis Failed

Error Log:

```
PiperTTS Failed to synthesize text: {e}  
Speech synthesis failed. Check whether the TTS model is available.
```

Cause:

- The TTS engine was not initialized correctly?
- The online speech service is unavailable?

Solution:

1. Verify that the TTS engine has been initialized correctly.
 2. Review the specific error message to take targeted corrective actions.
-

1.4 Errors Related to Serial Communication in the Voice Wakeup Module

Error 1.4.1: Failed to Open Serial Port

Error Log:

```
[kws_mic] Failed to open serial {port}: {e}  
Cannot open serial port {port}, please check voice wakeup module connection
```

Cause:

- The serial device for the voice wakeup module does not exist.
- Hardware connection issues.

Solution:

1. Check if the serial device exists: `ls -l /dev/mic`
 2. Verify that the hardware connections are secure.
-

1.4.2: Serial Port Reconnection Failure

Warning Logs:

```
[kws_mic] Max reconnect attempts ({max_attempts}) reached, giving up  
[kws_mic] Reconnect attempt {attempt} failed: {e}
```

Cause:

- The hardware device has disconnected.

Solution:

1. Reconnect (unplug and replug) the USB device.
-

1.4.3: Serial Port Transmission Failure

Error Logs:

```
send fail: {e}
```

Cause:

- The serial port connection has been severed.
- Buffer overflow.
- The voice wake-up device is unresponsive.

Solution:

1. Check the status of the serial port connection.
 2. Restart the program.
 3. Verify that the hardware device is functioning correctly.
-

1.5 ROS Communication-Related Errors

Error 1.5.1: Action Server Unavailable

Warning Logs:

```
play_audio action server not available, skip wakeup voice  
play_audio action server not available, skip error audio  
play_audio action server not available, skip voice
```

Cause:

- The `common_service` node has not been started.

Solution:

1. Confirm that the `common_service` node is running.
 2. Use `ros2 action list` to check for available actions.
-

Error 1.5.2: Action Goal Rejected

Warning Logs:

```
play_audio goal rejected  
TTS play goal rejected
```

Cause:

- The specified audio file path does not exist.

Solution:

1. Verify the existence of the audio file.
 2. Check the logs for the specific reason why the goal was rejected.
-

1.6 Large Language Model (LLM) Service-Related Errors

Error 1.6.1: Dify Connection Failure

Error Logs:

```
Dify LLM connection failed, Please check whether DIY dify has been started, API
and BASE_URL settings
Error log: {error_message}
Dify service connection failed: {status_code}
Dify service connection failed: {str(e)}
```

Cause:

- The Dify service has not been started.
- BASE_URL Configuration Error
- **Invalid API Key** (Most frequent cause: The API Key used in the Dify application does not match the `DIFY_API_KEY` specified in the configuration file.)

Solution:

1. Confirm that the Dify service has started and is running normally.
2. Verify the `DIFY_BASE_URL` configuration (e.g., `http://localhost/v1`).
3. Verify that the `DIFY_API_KEY` is correct.
4. Test the connection using `curl`: `curl -H "Authorization: Bearer {API_KEY}" {BASE_URL}/meta`

Error 1.6.2: Model Request Failed

Error Log:

```
The model request failed. Check whether the dify or AI model is normal.
Error Log: {error_message}
```

Causes:

- Dify service anomaly
- Model service outage
- Insufficient API credits for the model
- Request timeout

Solution:

1. Check the status of the Dify service.
 2. Review the Dify service logs.
 3. Check the API account status.
 4. Restart the Dify service.
-

1.7 Action Control-Related Errors

Error 1.7.1: TF Transform Retrieval Failed

Error Log:

```
Failed to get base_link-roi transform after {max_attempts} attempts; check if the TF tree is normal?
```

Cause:

- The TF tree is incomplete.
- The `pc1_segment` node is not publishing TF data.
- The target tracking module has lost the target.

Solution:

1. Use `ros2 run tf2_tools view_frames` to visualize the TF tree.
 2. Verify that the `pc1_segment` node is running correctly.
 3. Ensure that the node is actively tracking a target.
-

Error 1.7.2: TF Timestamp Too Old

Warning Log:

```
tf too old for base_link-roi: {age:.3f}s ago (attempt {attempt}/{max_attempts})
```

Cause:

- The TF publishing frequency is too low.
- System latency is too high, or CPU load is excessive.
- The target tracking module has lost the target.

Solution:

1. Increase the TF publishing frequency.
 2. Optimize system performance.
 3. Increase the value of the `tf_tolerance` parameter.
 4. Shut down unnecessary nodes.
-

Error 1.7.3: Joint Values Uninitialized

Warning Log:

```
joint1, joint_2, joint_4 values not initialized; skipping gaze control.
```

Cause:

- The robotic arm query service call failed.
- The initialization process was not completed.

Solution:

1. Check if the robotic arm service is functioning correctly.
2. Verify that the `arm_util` node has started.

3. Review the service call logs.

1.8 Audio Playback-Related Errors

Error 1.8.1: Wake-up Audio Loading Failed

Error Log:

```
Failed to load the wake-up word audio file: {e}
```

Cause:

- The audio file directory does not exist.
- The file format is not supported.

Solution:

1. Check the `system_voice/zh/wakeup` or `system_voice/en/wakeup` directory.
2. 3. Verify that the WAV file exists and is in the correct format.
3. Reinstall the `multi_brains_pre` package.

II. `waste_classify` Module

2.1 Model Loading Errors

Error 2.1.1: Model File Not Found

Error Log:

```
Model file not found: {model_path}
```

Cause:

- Incorrect configuration of the YOLO model path.
- The file has been deleted or moved.

Solution:

1. Check the configuration of the `model_path` parameter.
2. Verify that the model file exists.
3. Re-download the YOLO model file from the attached source code.

Error 2.1.2: YOLO Model Initialization Error

Error Log:

```
YOLO Model initialization error: {e}
```

Cause:

- The factory-default Python environment has been corrupted.
- Incorrect model file format.
- Insufficient memory.

Solution:

1. Verify that the model file is in a valid ONNX format.
 2. Attempt to use CPU mode: Set the `device` parameter to "cpu".
 3. Check if the system has sufficient memory.
-

Error 2.1.3: Model Inference Startup Failure

Error Log:

```
Model inference failed to start: {str(e)}
```

Cause:

- Shared memory creation failed.
- Multiprocessing startup failed.
- Insufficient resources.

Solution:

1. Check the system's shared memory limits: `sysctl kernel.shmmax`.
 2. Check the system logs for detailed error information.
-

2.2 Image Detection Errors

Error 2.2.1: Image Shape Not Initialized

Error Log:

```
Image shape not initialized; please ensure the camera is running and publishing images.
```

Cause:

- The camera has not started.
- The image topic is not being published.

Solution:

1. Verify that the camera node has started.
 2. Check if the topic `/camera/color/image_raw` contains data: `ros2 topic echo /camera/color/image_raw`.
 3. Restart the CarBase launch file
-

Error 2.2.2: Image Save Failed

Error Log:

```
Failed to save image: {e}
```

Cause:

- The output directory does not exist.
- Insufficient disk space.

- The camera node has not started.

Solution:

1. Check the file path configuration.

III. pcl_seg Point Cloud Segmentation Module

3.1 Depth Image Processing Errors

Error 3.1.1: Invalid ROI Region (Common)

Warning Log:

```
Invalid ROI Region: [umin vmin umax vmax]
```

Cause:

- Tracking box coordinates exceed image boundaries
- Tracking box dimensions are zero or negative
- The target region lacks depth information

Solution:

1. Check if the tracking box coordinates are valid/reasonable
2. Use RViz to verify whether depth information exists for the target

Error 3.1.2: No Depth Image

Warning Log:

```
No depth image received.
```

Cause:

- The depth camera has not been started
- Topic subscription failed
- The depth image is not being published

Solution:

1. Confirm that the depth camera driver has been started
2. Check if the topic `/camera/depth/image_raw` contains data
3. Verify the camera hardware connections
4. Restart the camera node

Error 3.1.3: No Valid Depth Points in Circular Region

Warning Log:

```
No valid depth points found in the circular region.
```

Cause:

- The target point is too far or too close; there is no depth information in the vicinity of the target point
- Depth values are invalid (0 or exceed the defined threshold)

Solution:

1. Adjust the camera's viewing angle or the position of the target point
 2. Check if the depth camera is functioning correctly
 3. Use RViz to verify whether depth information exists for the target
 4. Reduce ambient lighting
-

Error 3.1.4: Target Point Outside Image Boundaries

Warning Log:

The target point is outside the image boundary. Image size: {width}x{height}, Requested: ({x}, {y})

Cause:

- The requested coordinates fall outside the image range
- Coordinate calculation error
- Image resolution has changed

Solution:

1. Verify the coordinate calculation logic
 2. Check the image resolution configuration
 3. Implement boundary checks
 4. Use valid coordinate values
-

Error 3.1.5: Invalid Radius Value

Warning Log:

Invalid radius value.

Cause:

- The radius parameter is negative or zero.
- Parameter passing error.

Solution:

1. Ensure the radius value is a positive number.
 2. Check the service call parameters.
 3. Use a reasonable radius value (5–50 pixels recommended).
-

3.2 TF Transform-Related Errors

Error 3.2.1: TF Transform Unavailable

Warning Log:

```
TF transform not available between base_link and camera_color_optical_frame.
```

Error Log:

```
Failed to lookup transform: {ex.what()}
```

Cause:

- Necessary transforms are missing from the TF tree.
- The camera coordinate frame has not been configured.
- The TF publishing node has not been started.

Solution:

1. Use `ros2 run tf2_tools view_frames` to inspect the TF tree.
 2. Ensure that the camera driver is publishing the correct TF transforms.
 3. Verify that the transform from `base_link` to the camera is correctly defined.
-

Error 3.2.2: TF Transform Failed

Error Log:

```
Failed to transform actual_pose: {ex.what()}  
Failed to transform estimate_pose: {ex.what()}  
Failed to transform point: {ex.what()}
```

Cause:

- The specified coordinate frame does not exist.
- The transform data is invalid or corrupted.
- Timestamp synchronization issues.

Solution:

1. Verify that the coordinate frame names are correct.
 2. Validate the integrity of the TF data.
 3. Check for timestamp synchronization issues.
 4. Use `ros2 run tf2_tools echo` to inspect the TF data.
-

3.3 Point Cloud Processing-Related Warnings

Error 3.3.1: ROI Point Cloud is Empty

Debug Log:

```
ROI cloud is empty, skipping this depthframe
```

Cause:

- There are no valid depth values within the specified ROI region.
- The distance threshold parameter is set too low.
- The target object is outside the valid distance range.

Solution:

1. Increase the `distance_threshold` parameter.
 2. Adjust the size of the ROI region.
 3. Ensure that the target object is within the valid distance range.
 4. Check the quality of the depth image data.
-

Error 3.3.2: No Cluster Found**Debug Log:**

```
No cluster found  
No suitable cluster found
```

Cause:

- The point cloud density is insufficient.
- Clustering parameters have been configured incorrectly.
- Target is too small or too distant

Solution:

1. Decrease the `cluster_min_size` parameter.
 2. Increase the `cluster_max_size` parameter.
 3. Adjust the `clusterTolerance` parameter.
 4. Ensure the target is sufficiently close and clearly visible.
-

Error 3.3.3: No Valid Center Data**Warning Log:**

```
GetBoxCenter: no valid data.
```

Cause:

- Depth images have not yet been processed.
- Clustering failed.

Solution:

1. Ensure that depth images are being processed correctly.
 2. Check if point cloud segmentation was successful.
-

IV. super_tracker Target Tracking Module

4.1 Tracker Initialization Errors

Error 4.1.1: Tracker Creation Failed

Error Log:

```
Failed to create tracker: {e.what()}
```

Cause:

- The model file does not exist.

Solution:

1. Check the model file path: `$HOME/MODELS/tracker/mixformer_v2_sim.onnx`
 2. Adjust the `num_threads` parameter.
-

Error 4.1.2: Display Window Configuration Conflict

Error Log:

```
display_windows and display_windows_only_track cannot both be true. Setting both to false.
```

Cause:

- Both display parameters were set to `true` simultaneously.
- Configuration logic conflict.

Solution:

1. Enable only one of the parameters:
 - `display_windows`: Always display the window; supports mouse selection.
 - `display_windows_only_track`: Display the window only when tracking.
 2. Modify the configuration file to ensure that both are not set to `true` at the same time.
-

4.2 Image Frame Processing Warnings

Error 4.2.1: Received Empty or Invalid Frame

Warning Log:

```
Received empty or invalid frame from camera
```

Cause:

- The camera is not publishing images correctly.

Solution:

1. Check the running status of the camera node.
2. Use `ros2 topic hz` Check the publishing frequency of `/camera/color/image_raw`

4.3 ROI Selection Errors

Error 4.3.1: Selecting ROI on an Empty Frame

Error Log:

```
Cannot select ROI on empty frame
```

Cause:

- No image frames have been received yet.
- The image frame is empty.

Solution:

1. Wait for the camera to publish images normally before selecting the ROI.
 2. Check the camera connection.
 3. Verify that the image topic contains data.
-

Error 4.3.2: Invalid ROI Selection

Warning Log:

```
Invalid ROI selected
```

Cause:

- The user cancelled the ROI selection.
- The selected ROI has zero or negative width/height dimensions.

Solution:

1. Re-select a valid ROI area.
 2. Ensure the ROI area has sufficient dimensions (a minimum of 20x20 pixels is recommended).
 3. Enclose the target completely within the bounding box on the image.
-

Error 4.3.3: Failed to Set Tracking ROI

Error Log:

```
Invalid frame when setting track ROI  
Failed to Set Track ROI {e.what()}
```

Cause:

- The image frame is empty.
- The tracker initialization failed or encountered an exception.
- The ROI coordinates are invalid.

Solution:

1. Ensure that a valid image frame is provided.
2. Check if the ROI coordinates fall within the image boundaries.
3. Verify the status of the tracker.

4. Re-initialize the tracker.

Error 4.3.4: Tracking Startup Failed

Error Log:

```
Failed to start tracking
```

Cause:

- The current camera frame is invalid.
- The ROI setting failed.
- The tracker encountered an exception.

Solution:

1. Confirm that the camera image is normal.
2. Check the validity of the ROI coordinates.
3. Review the specific error logs for details.

V. General Troubleshooting Suggestions

5.1 Adjusting Log Levels

- If you encounter an issue, enable debug logging mode and send the complete error output to technical support.
- ROS Log Level: `ros2 launch xxxx --ros-args --log-level debug`

5.2 Common Diagnostic Commands

```
# View list of topics
ros2 topic list

# View topic data
ros2 topic echo <topic_name>

# View list of services
ros2 service list

# View list of Actions
ros2 action list

# View TF tree
ros2 run tf2_tools view_frames

# Check node status
ros2 node list

# View logs
ros2 launch xxxx --ros-args --log-level debug
```

